Requirements Document

Topic 14: Switch Node & Workflow Routing

1. Introduction

This document describes the functional and technical requirements for implementing a role-based routing workflow using the Switch node in n8n.

2. Course Objective

The objective is to teach trainees how to configure the Switch node to route workflow items through multiple output branches based on specified conditions.

3. Scope

3.1 In Scope

- Creating a new n8n workflow.
- Configuring mock user profiles.
- Splitting an array into individual workflow items.
- Configuring the Switch node.
- Creating multiple comparison rules.
- Using dynamic expressions.
- Configuring a fallback output.
- Connecting downstream Edit Fields nodes.
- Testing multiple input items.

3.2 Out of Scope

- Custom JavaScript switch statements in Code nodes.
- Runtime-generated output branches.
- External database integration.
- Authentication and authorization enforcement.

4. Functional Requirements

ID	Requirement	Description

REQ-001	Switch Integration	The workflow must contain a Switch node.
REQ-002	Multiple Rules	The Switch node must support admin, editor, and viewer rules.
REQ-003	Fallback Route	Unmatched roles must be routed through a fallback output.
REQ-004	Rule Ordering	Routing rules must be configured in the specified top-to-bottom order.
REQ-005	Dynamic Expression	The workflow must evaluate the role using `{{ $json.role }}`.
REQ-006	Downstream Nodes	Four separate Edit Fields nodes must handle the four outputs.
REQ-007	Multiple Items	The workflow must process multiple user profiles.

5. Technical Requirements

ID	Requirement
TR-001	The Switch node must route items to the appropriate output based on matching conditions.
TR-002	The workflow must split multiple JSON profiles into separate items before routing.

6. Input Requirements

The workflow must accept user profile information containing:

- name
- role

Supported sample roles:

- admin
- editor
- viewer
- guest

7. Output Requirements

Output	Destination Node	Required Status
0	Set Admin	Admin Access

1	Set Editor	Editor Access
2	Set Viewer	Viewer Access
3	Set Guest	Guest Access

Each downstream node must preserve the original user information and add a string field named `status`.

8. Assumptions and Constraints

Assumptions

- Each input profile contains a role field.
- The workflow receives valid JSON data.
- The role values are strings.
- The guest role is handled by the fallback route.

Constraints

- The number of Switch outputs is configured in advance.
- Routing depends on the configured conditions.
- The workflow uses n8n's built-in nodes.

9. Acceptance Criteria

The project is considered successful when:

1. The Switch node is configured correctly.
2. All three role conditions are configured.
3. The fallback output is enabled.
4. Four downstream nodes are connected.
5. Multiple user profiles are split into individual items.
6. The workflow executes successfully.
7. Each user is routed to the appropriate branch.